

CHANDAN SINGH

AI/ML Engineer - Computer Vision - NLP - Full-Stack AI Systems

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PROFESSIONAL SUMMARY

B.Tech CSE student (Expected 2027, CGPA 8.3/10 through 6th semester) specializing in AI/ML engineering with hands-on experience building deployable computer vision, NLP, and full-stack AI systems. Shipped 8 end-to-end projects spanning YOLOv8, LSTM, XGBoost, SegFormer/DeepLabV3+, and LLM-based agentic pipelines, competing in national and AMD-sponsored hackathons.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL

AI/ML Frameworks: TensorFlow, PyTorch, scikit-learn, segmentation-models-pytorch, SHAP, MediaPipe

Backend & Cloud: FastAPI, Flask, Docker, Git, HuggingFace Hub, ROCm, Alembic

Full-Stack & DB: React, Vite, PostgreSQL, SQLAlchemy, JWT Authentication, Twilio API

Generative AI & Agents: LangChain, RAG, FAISS, Jina/Vector Embeddings, Groq API, Agentic Pipelines

Concepts: Computer Vision, NLP, Semantic Segmentation, Predictive ML

TECHNICAL PROJECTS

NikaAI - Industrial Defect Detection & Quality Intelligence Platform | YOLOv8s - FastAPI - Celery - Vercel

- Engineered a production-grade computer vision platform for the AMD AI Hackathon; fine-tuned YOLOv8s achieving 83% mAP@0.5 (56.4% mAP@0.5:0.95) across 17 defect classes on 29,354 industrial images, deployed via FastAPI, Celery, and Vercel.
- Implemented Monte Carlo Dropout uncertainty quantification (T=30 passes) to flag low-confidence predictions, plus a Gemma LLM reasoning layer for automated root-cause analysis, secured with JWT-based RBAC.
- Published the underlying detector to HuggingFace Hub (negi3961/factory-defect-guard) as a standalone, citable model artifact for reproducibility.

HR-RAG-Assistant - Retrieval-Augmented HR Policy Q&A | LangChain - FAISS - Jina Embeddings - Groq

- Built a full RAG pipeline answering natural-language queries over HR policy documents via a LangChain agent tool with a Streamlit chat UI.
- Architected a modular pipeline (loader, chunker, embeddings, vector store, agent) with a persisted FAISS index for instant reloads and tunable chunk size / top-k retrieval.

MedVoice-AI - Agentic Clinical Documentation Backend | FastAPI - OpenAI/Sarvam - HuggingFace - Groq

- Developed a FastAPI backend for outpatient clinical documentation - visit records, audio transcription, EMR generation, consent logging, and PDF export.
- Designed a bounded agentic decision layer on Groq's API (transcription-quality gate, EMR self-check, triage agent) with fail-safe defaults for judgment-based checks.

HABS - Healthcare Appointment Booking System | FastAPI - React - Random Forest - SHAP - Alembic

- Built an async SQLAlchemy backend (6-table schema, Unit-of-Work pattern, double-booking guard) with a Random Forest no-show prediction model and SHAP-based explainability.
- Designed a full-stack booking flow end-to-end - React frontend, FastAPI backend, Alembic migrations - supporting doctor scheduling and ML-driven no-show flags.

OTHER PROJECTS

TRINETRA - AI Scam Intelligence Engine (XGBoost - Twilio - Google Safe Browsing - team project) - Contributed to a real-time fraud detection platform (95%+ accuracy) analyzing URLs, SMS, and WhatsApp screenshots, with a 0-100 threat score and Twilio WhatsApp bot.

SIGNSPEAK - Word-Level Sign Language Recognition (LSTM - MediaPipe - React/Vite - Flask - team project) - Contributed to a real-time sign recognition system detecting complete words via 30-frame LSTM sequences over MediaPipe hand landmarks, shipped across web, desktop, and API interfaces.

Falcon Offroad Semantic Segmentation (DeepLabV3+ - mit_b2 (SegFormer) - Albumentations) - Built an offroad segmentation model for HackWithMumbai 2.0 on Duality AI's Falcon desert digital-twin dataset, achieving 72% mIoU with a custom Focal + Dice loss.

ACHIEVEMENTS & HACKATHONS

- AMD AI Hackathon - designed and deployed NikaAI, a production-oriented computer vision platform, in a hackathon track sponsored with a \$20,000+ overall prize pool.
- HackWithMumbai 2.0 - built the Falcon offroad semantic segmentation model on Duality AI simulation data as part of the BWI-PS track.
- Self-directed model publishing - released a fine-tuned defect-detection model on HuggingFace Hub for public reuse and reproducibility.

EDUCATION

B.Tech, Computer Science & Engineering - Expected 2027

Maharishi University of Information Technology, Uttar Pradesh, India | CGPA: 8.3 / 10 (through 6th semester)